



Network Tariff Report 2002

1st March 2002

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1 INTRODUCTION

As part of its *Electricity Distribution Price Determination*, the Office of the Regulator General (the Office), now known as the Essential Services Commission (the Commission) required the Distributors prepare a annual report on their network tariffs. This requirement is detailed in clause 2.3.13 of *Electricity Distribution Price Determination - Volume II* which provides that the report should include (but not be limited to):

- the distribution tariffs and transmission tariffs charged by the Distribution business;
- a discussion of the policy framework and tariff principals adopted by the distribution business in framing the structures and level of its tariffs;
- an explanation of the basis on which the distribution business has determined the appropriate breakdown between fixed and variable charges;
- the rationale for the introduction of any new tariffs, or the withdrawal of tariffs;
- an explanation of how the distribution business has had regard to the consideration of upper and lower bounds contained in clause 2.3.12 in determining its distribution tariffs;
- the extent to which the tariff structures adopted provide efficient consumption signals to distribution customers;
- the methodology adopted for allocating transmission related costs to distribution customers through its transmission tariffs, as submitted to the Office in line with clause 3.1.10; and
- an estimate of the average annual distribution and transmission charge (in \$) for each combination of distribution and transmission tariff.

The Office may from time to time issue further guidance on the form and scope of the information contained in the Tariff Report. The distribution businesses will be required to comply with any such guidance issued by the Office

2 OVERVIEW OF NETWORK TARIFF COMPONENTS

Network tariffs consist of two components:

- distribution related charges (DUoS) to recover the regulated revenues for the distribution business; and
- transmission related charges (TUoS) to recover the costs of transmission use of system services provided by Victorian Energy Networks Corporation (VENCorp), transmission connection services provided by SPI PowerNet, equalisation payments prescribed by regulation and inter distribution system charges.

These various components are described in detail in Sections 2.1 to 2.3 below.

2.1 *Distribution Use of System Charges (DUoS)*

Leading up to each five year regulatory period the Commission determines the revenue requirement for each distribution business over the next regulatory period. The Office has utilised a building block methodology to determine the amount of required revenue for the purposes of the previous price review. Recovery of the aggregate distribution revenue requirement determined by the Commission is achieved through the application of DUoS tariffs. In terms of distribution prices from 2001 to 2005, the final outcome of the Distribution Price Review for CitiPower was a real reduction in average distribution prices of 11.2% in 2001 and 1% in subsequent years. This 1% reduction is reflected in the approved 2002 tariffs.¹

The application of distribution tariffs is governed by the price controls established by the Office, and contained in the Electricity Distribution Price Determination 2001-2005 – Volume II. The key provisions of this document define:

- the price control mechanisms applied to determine whether proposed prices are compliant; and
- the constraints on adjusting existing tariffs, called re-balancing constraints.

2.1.1 The Price Control

The Office has introduced a tariff basked approach to regulate Distribution prices which is essentially a CPI-X price control based on tariff components weighted by their respective component quantities. The price regulation mechanism is based entirely on actual sales and does not rely on forecasts.²

- The maximum $(1+CPI_t)*(1-X_t)$ ratio allowed for 2002 is 1.03338 based on:-
- All Groups Index, Weighted Average of Eight Capital Cities, September quarter 2001 of 134.2
- All Groups Index, Weighted Average of Eight Capital Cities, March quarter 2000 of 130.9
- X_t for 2002 of .01
- GST inflationary adjustment of 0.011
- Adjustment factor of (0.727%) for the Distribution revenues relating to the financial year 1999/2000 and the six months to December 2000.

The calculation includes an adjustment to distribution prices in 2002 to correct for residual adjustments necessary from the pre December 31, 2000 regulatory period.

2.1.2 Rebalancing Constraints

¹ CitiPower's network tariff schedules are contained in Appendix 1 of this document.

² For a comprehensive description of the price controls see Office's "Electricity Distribution Price Determination 2001-2005, Volume II – Price Controls"

The average price of a network tariff is limited by the re-balancing constraint. This control effectively limits the rate of increase of individual network tariffs. The re-balancing constraint for 2002 for both transmission and distribution tariffs is CPI plus 2%.

2.2 *Inter Distribution System Charges*

The CitiPower distribution network is interconnected with the neighbouring distribution businesses of AGL and United Energy. These interconnected DBs buy and sell distribution capacity across their distribution boundaries. This practice is efficient and avoids the unnecessary duplication of assets.

CitiPower is a net purchaser of inter-DB capacity and under the existing pricing regulations these net costs are recovered through the Network Tariffs.

CitiPower has assumed a net cost of \$1.9 million per annum for inter distribution system services. These costs are incorporated into the Network Tariffs in the same way as transmission charges.

2.3 *Transmission Charges*

Costs associated with the use of and the connection to the Transmission system are incurred by CitiPower and incorporated into the transmission component of CitiPower's Network Tariffs to be passed through to distribution customers.

The elements of TUoS are: Prescribed Connection Charges, Excluded Transmission Services, Transmission Use of System charges, Settlement Residues, Equalisation and Inter Distribution System Charges. For a more detailed description of each of these elements please refer to CitiPower's Distribution Price Review Submission of December 1999.

3 PRICING METHODOLOGY FOR TRANSMISSION CHARGES

The transmission charges which CitiPower pays to VENCORP and SPI PowerNet consist of components that can be separated into demand based and energy based components. These components are then recovered on the basis of "user pays", as set out below, to derive the TUoS price components of the Network Tariffs.

The following elements of the Transmission Charges were assigned to the Demand component:

- Connection charges;
- Maximum Demand Charges;
- Summer Peak Energy Charges;
- Excess Reactive Demand Charges; and
- Settlement residues (Offset against transmission charges).

The remaining elements listed below were allocated to the Energy Component:

- VENCORP Common Service Charge (\$/MW on all energy transmitted); and
- Equalisation Adjustment (a tax which should be recovered in a manner that gives rise to the least possible distortion of economic signals).

From these components a demand and energy price were calculated.

With reference to experience in the Network Planning group and from the End Use Measurement Survey, it was possible to arrive at a series of load factors and peak contribution factors which were used to allocate transmission costs to the tariff groups in proportion to the demand presented at the time of CitiPower's system peak and the total energy consumed on the basis of the demand and energy prices derived above.³

3.1 Transmission Charges

The transmission charges are then established to collect the required revenue on the basis of available metering data. For example, the price for single rate customers is derived by dividing the revenue required by the forecast energy. In the case of two rate customers, the off peak consumption was assigned the Transmission energy price and the peak consumption was used to recover the balance of the required revenues.

In CitiPower's opinion there is no compelling reason for a customer standing charge to be applied for the recovery of transmission costs. For residential customers it can be assumed that there is a reasonable correlation between energy usage and contribution to the local and system network maximum demand. Network tariffs for larger customers contain a demand related component which in part reflects the cost of transmission system usage. Therefore there is no recovery of transmission charges through the standing charge component of network tariffs.

Energy components of transmission charges such as the Common Service Charge and the Equalisation payment have been allocated to the off-peak component. The concept of applying the Equalisation charge and VENCORP common service charge on the basis of energy use results in a higher allocation of transmission charges to off-peak periods. To manage the impact of this on large customers with high off-peak consumption, this change will be transitioned to so as to avoid sudden impacts on customers. This is not expected to result in off-peak transmission charges rising to uneconomically high levels (ie above stand alone costs). It is worth noting that TUoS has the potential to be volatile from year to year due to seasonal effects and the very volatile settlements residue.⁴

³ The End Use Measurement Study commenced in 1994 as a Victorian Distribution Industry load monitoring study. Smart meters compile load data on a sample of residential, commercial and industrial customers. The study is ongoing and CitiPower continue to participate.

⁴ The operation of the wholesale market results in a settlement surplus arising from interconnector constraints and the application of marginal loss factors. The National Electricity Code requires that these settlement surpluses be returned to market participants through reduced transmission charges. The settlement surplus is largely determined by the difference in regional pool prices at the time when the interconnector becomes constrained. It is therefore very volatile.

4 ECONOMICALLY EFFICIENT DISTRIBUTION PRICES

Clause 2.3.12 of the *Electricity Distribution Price Determination - Volume II* requires the distribution businesses to have regard to the upper and lower bounds in setting distribution prices, and therefore satisfy the criteria for economic efficiency. These upper and lower bounds are defined as follows:

1. each distribution tariff should be above the avoided cost of serving distribution customers assigned to that distribution tariff; and
2. each distribution tariff should be below the cost of providing the service on a stand alone basis to distribution customers assigned to that distribution tariff.

As part of the Electricity Distribution Price Review process CitiPower engaged consultants Ernst & Young to address this issue and they provided an assessment of the *future costs* and *stand-alone cost* of distribution services in order to demonstrate that CitiPower's tariffs fall within the bounds of economically efficient prices. This report is attached to CitiPower's Distribution Price Submission as Appendix 7 .

The Ernst & Young report and the approach taken in the development of their report is discussed in CitiPower's 2001 Electricity Distribution Price Review Submission to the Office of the Regulator General – 1st December 1999. The conclusions of this report have been updated (below) to reflect CitiPower's approved 2002 distribution tariffs.

Table 1 below summarises the results for the calculation of the cost window for each tariff category. Three stand-alone costs have been assessed - for small users; LV demand metered users; and HV users. Each tariff has been attributed a stand-alone cost according to whether it falls within these categories. This has necessarily involved a degree of averaging as well as the use of broad-brush assumptions. However, the results are robust both in terms of the approach that has been adopted and the numerical outcomes as shown in the following table.⁵

The range index shows the relative position of the average distribution price is within the range between avoidable cost and stand-alone cost. A range index value of 50% indicates that the average price lies at the mid-point of the range.

⁵ Note that the numbers in Table 1 are in nominal dollars of the day, compared with the presentation in CitiPower's Distribution Price Review Submission which was based on real 1999 dollars.

TARIFF CATEGORY	Avoidable Cost (c/kW)	2002 Distribution Price (c/kWh)	Range Index	Stand alone Costs (c/kWh)
Residential single rate	1.8	4.7	52%	7.4
Residential single rate –Bulk	1.5	3.9	40%	7.4
Residential two rate	1.0	3.1	33%	7.4
Residential two rate –Bulk	1.1	2.7	25%	7.4
Dedicated supplies	0.3	1.7	20%	7.4
Dedicated supplies –Bulk	0.2	1.5	18%	7.4
General Purpose single rate	1.1	5.0	62%	7.4
General Purpose single rate -Bulk	0.9	4.1	49%	7.4
General Purpose two rate (5 day peak)	0.9	4.0	48%	7.4
General Purpose two rate (5 day peak) –Bulk	0.7	3.3	39%	7.4
General Purpose two rate (7 day peak)	1.0	4.2	50%	7.4
General Purpose two rate (7 day peak) –Bulk	1.0	3.3	36%	7.4
Large two rate (7 day peak) -Bulk	1.0	3.5	39%	7.4
Unmetered Supplies	0.4	2.6	31%	7.4
Large LV Demand metered	0.7	2.6	32%	6.7
HV Demand Metered	0.3	1.8	64%	2.7

Table 1: Economically Efficient Distribution Prices

Table 1 shows that the approved distribution tariffs satisfy the economic efficiency criteria.

The avoidable cost limit represents the lowest possible charge at which a commercial organisation would provide the service. At this level, there is sufficient revenue generated to enable the service provider to recover its marginal or avoidable cost, however there is no contribution to the service provider's fixed costs. Therefore, pricing on such a basis will not sustain a commercial business.

5 NETWORK TARIFFS

As required by the Office's Final Determination, CitiPower has produced a schedule of network, distribution and transmission tariffs. The network tariffs are a simple addition of the distribution and transmission components. These tariff schedules are available in appendix 1. These tariffs recover the distribution revenue requirement (as set out in the Office's Price Determination) plus the assumed transmission and inter distribution business charges.

CitiPower recognises that tariff design ultimately requires compromises to be reached in relation to a range of conflicting objectives such as:

- economic efficiency;
- energy efficiency;
- pricing stability;
- commercial imperatives; and
- customer equity.

CitiPower has considered, and will continue to consider, in an even-handed way, the trade-offs between these objectives, in formulating new tariffs or modifications to existing ones. These objectives are embodied in CitiPower's network charges, discussed in section 5 below.

5.1 *List of approved Network Tariffs*

Each of CitiPower's network tariffs are examined in the sections below. All prices and annual bill calculations exclude GST.

5.1.1 *Residential Single Rate Tariffs*

The Residential Single Rate and Residential Single Rate - Bulk tariffs were originally based on retail Maximum Uniform Tariffs (MUTs) for residential customers. The recently expired MUT for single rate residential customers includes an inverse block structure such that the price for the first 1020 kWh per quarter is slightly lower than for the balance.

This inverse block structure has been introduced to the residential network tariff to maintain a positive rudimentary demand management signal to larger than average consumers. The average consumption of CitiPower single rate residential customers is approximately 4300 kWh per annum which is close to the 1020 kWh per quarter (4080 kWh per annum) used in the MUTs GD and GR. For simplicity the same 1020 kWh transition point has been retained.

On the basis that residential customers, in aggregate, present a different demand profile to commercial and industrial customers these tariffs are restricted to residential customers. Figures 1 and 2 below illustrate diagrammatically the impact the significant reductions have had in delivering price reductions to small residential customers, and the block structure in providing a price incentive to large residential customers.

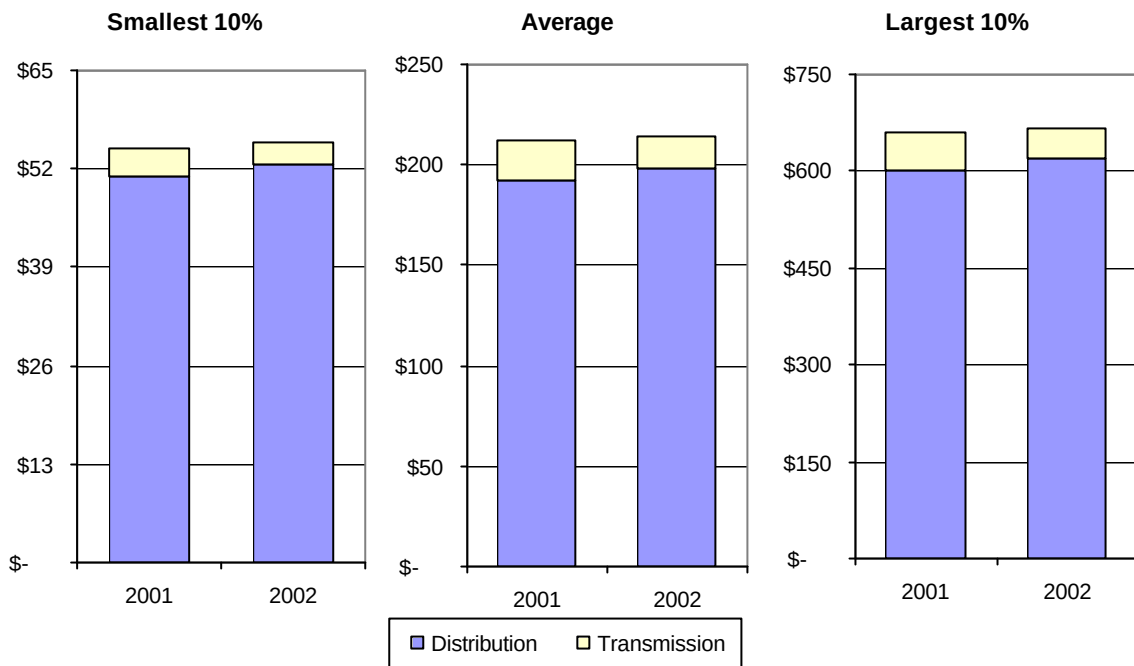


Fig 1: Residential Single Rate – Average Annual Bill by Customer Segment

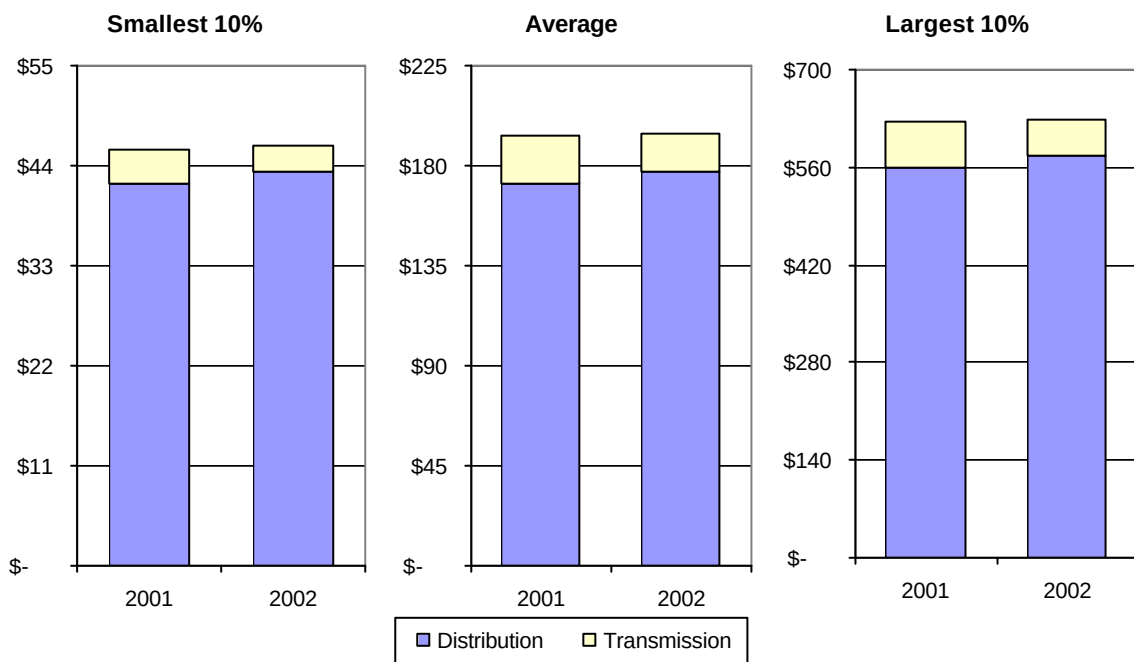


Fig 2: Residential Single Rate Bulk – Average Annual Bill by Customer Segment

5.1.2 Residential Two Rate Tariffs

The Residential Two Rate and Residential Two Rate - Bulk tariffs were originally also based on retail Maximum Uniform Tariffs (MUTs) for residential customers and include a rudimentary demand signal through the differential pricing for time of use.

On the basis that residential customers, in aggregate, present a different demand profile to commercial and industrial customers these tariffs are restricted to residential customers.

The impact of these tariffs is shown diagrammatically in Figures 3 and 4 below.

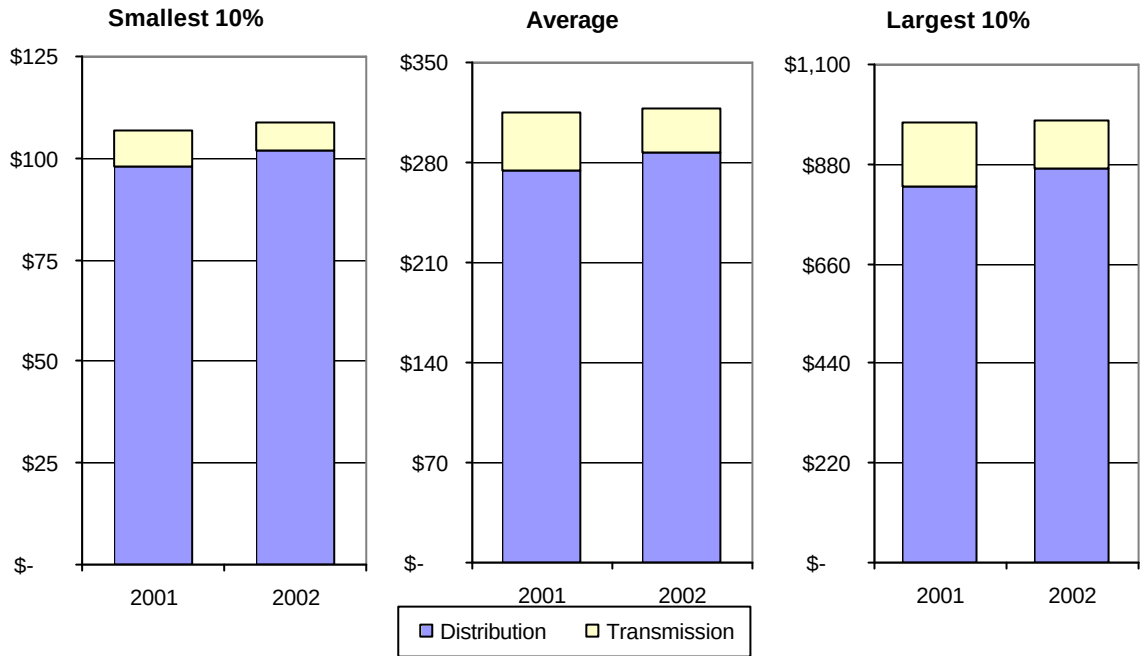


Fig 3: Residential Two Rate – Average Annual Bill by Customer Segment

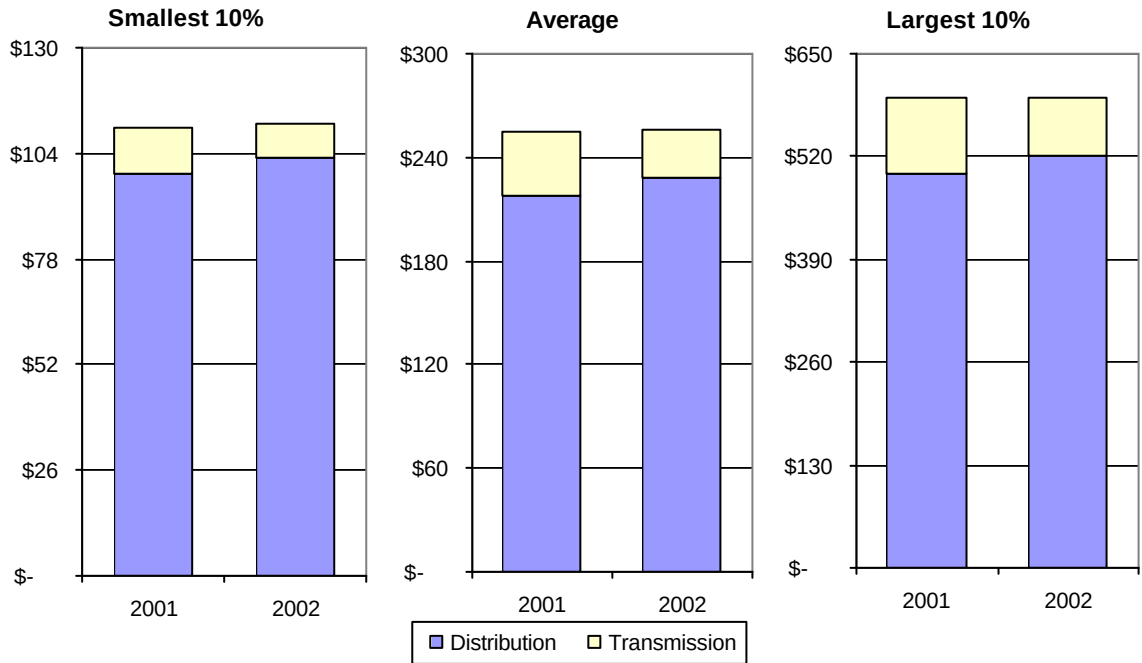


Fig 4: Residential Two Rate Bulk – Average Annual Bill by Customer Segment

5.1.3 Dedicated Supplies (Storage Water & Space Heating)

These tariffs are available for controlled loads such as storage water heating and space heating for residential purposes only. Being controlled loads, they make no contribution to peak demands on either the transmission or distribution system.

These tariffs currently make a substantial contribution to the total revenue without driving capacity increase while at the same facing intense competition from alternative fuel sources such as natural gas. Therefore the large reductions in the 2001 rates to make these tariffs more affordable have been maintained. This should help discourage these sales being unnecessarily eroded to the detriment of system utilisation and therefore overall economic efficiency of the distribution network.

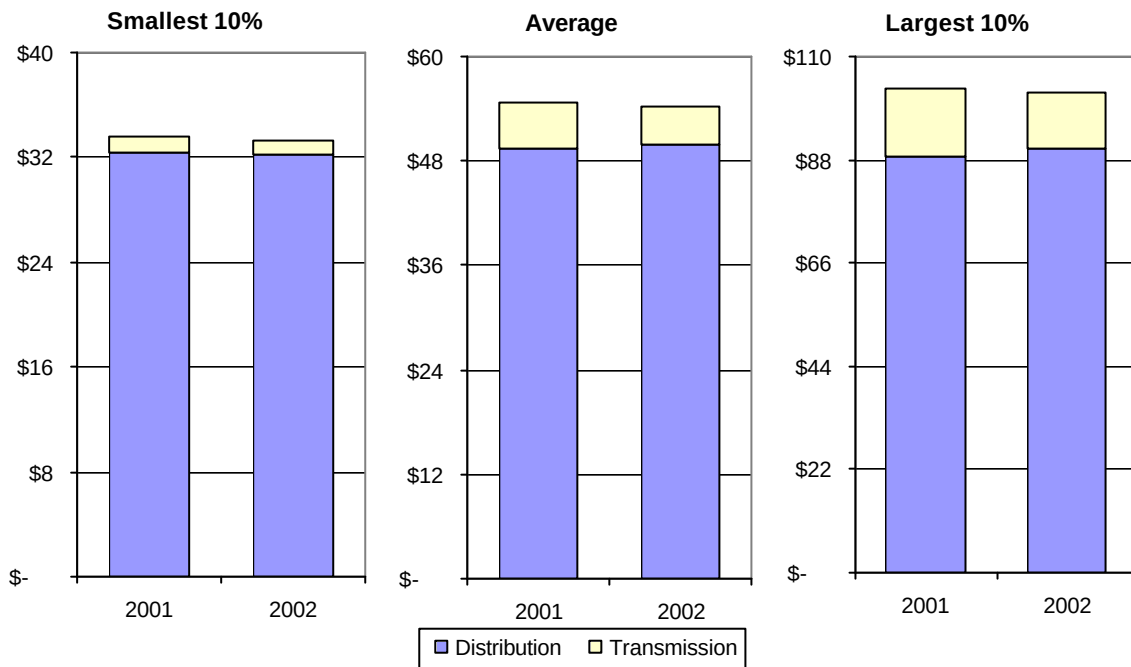


Fig 5: Dedicated Supplies – Average Annual Bill by Customer Segment

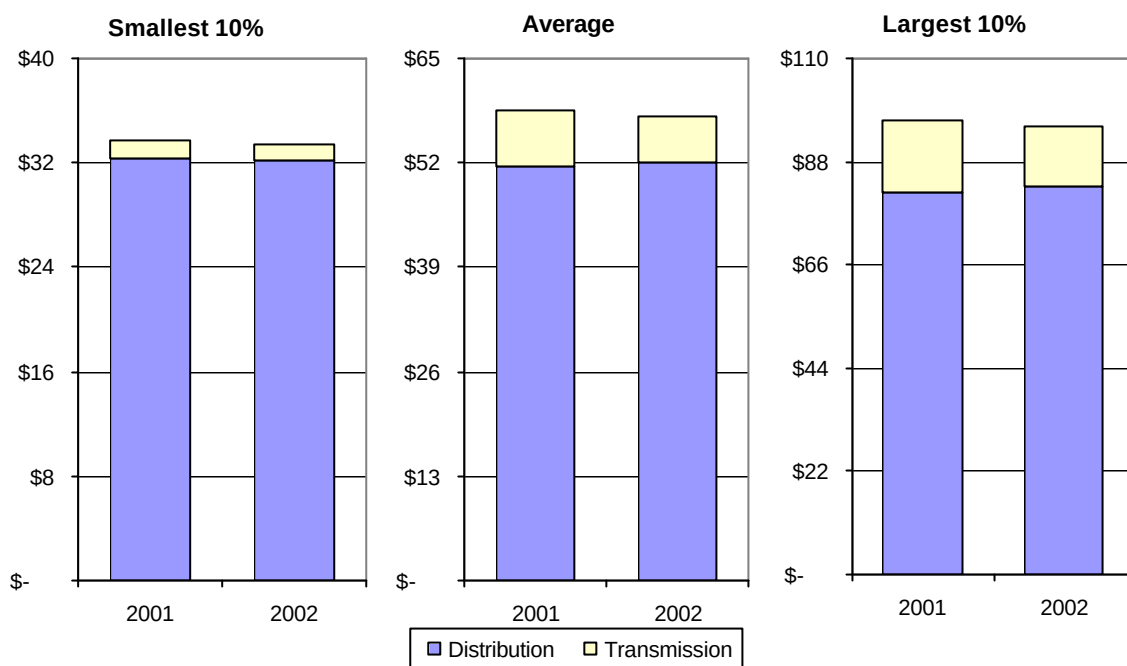


Fig 6: Dedicated Supplies Bulk – Average Annual Bill by Customer Segment

5.1.4 General Purpose Single Rate .

These tariffs were based on the single rate General Purpose MUT previously available to small and medium business and industrial customers.

No major changes have been made to the structure of these tariffs and new customers are encouraged to consider the two rate or maximum demand tariffs as an alternative, as they promote more efficient utilisation of the network by providing a time of use signal.

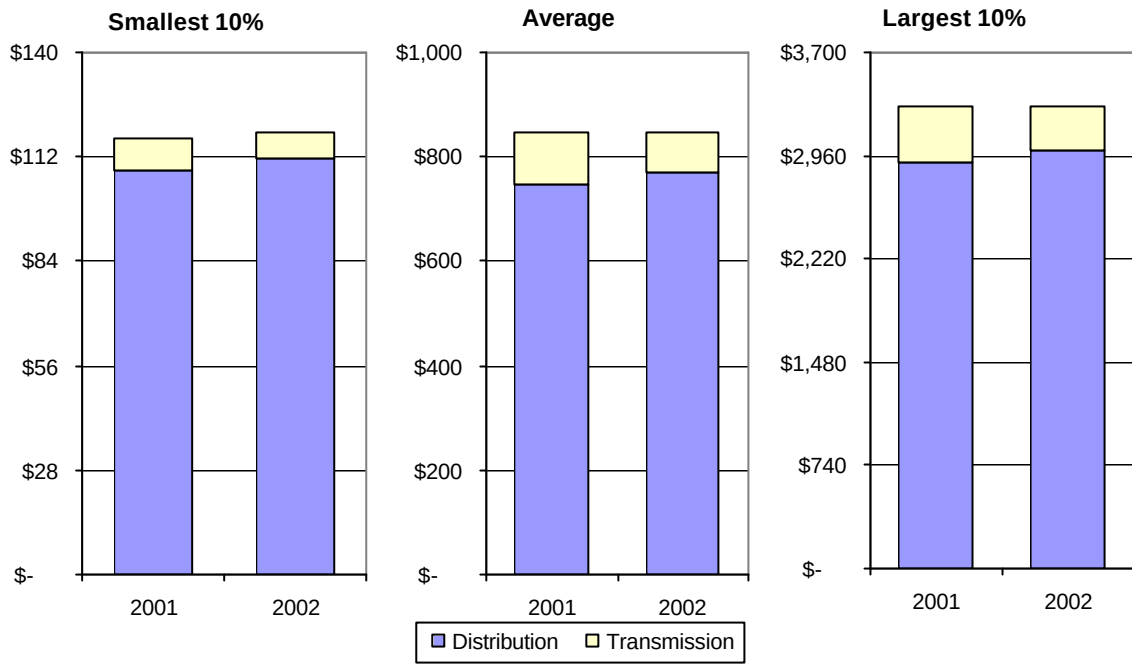


Fig 7: General Purpose Single Rate – Average Annual Bill by Customer Segment

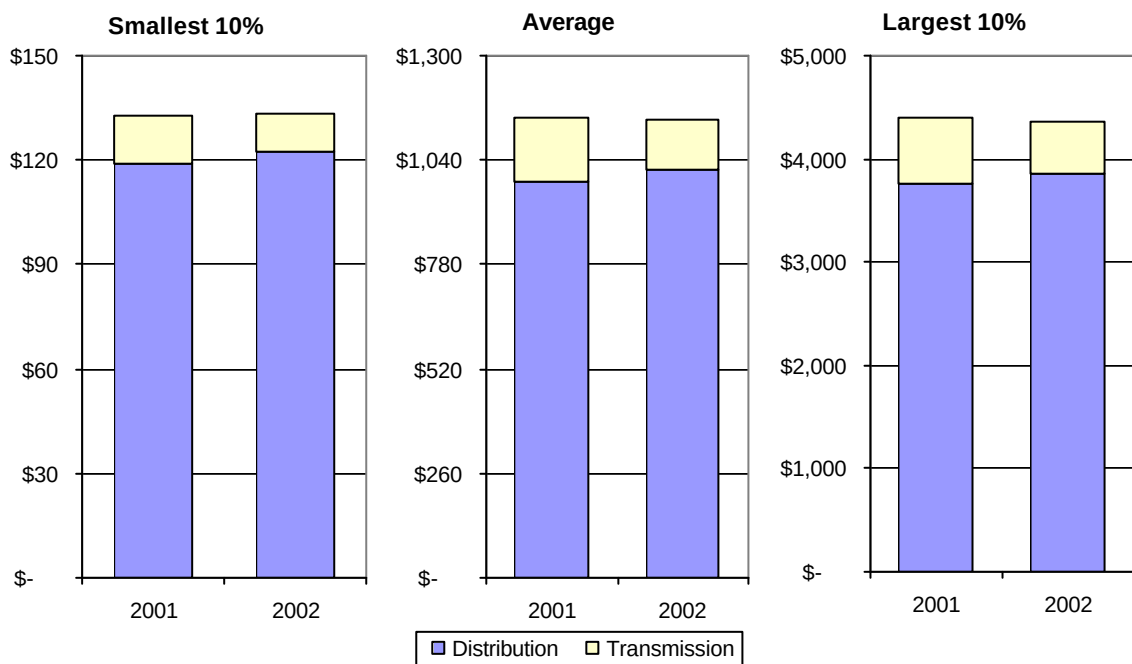


Fig 8: General Purpose Single Rate Bulk – Average Annual Bill by Customer Segment

5.1.5 General Purpose Two Rate (5 day peak) .

These tariffs were based on the two rate General Purpose MUT used by small and medium business and industrial customers and include a rudimentary demand signal through the differential pricing for time of use.

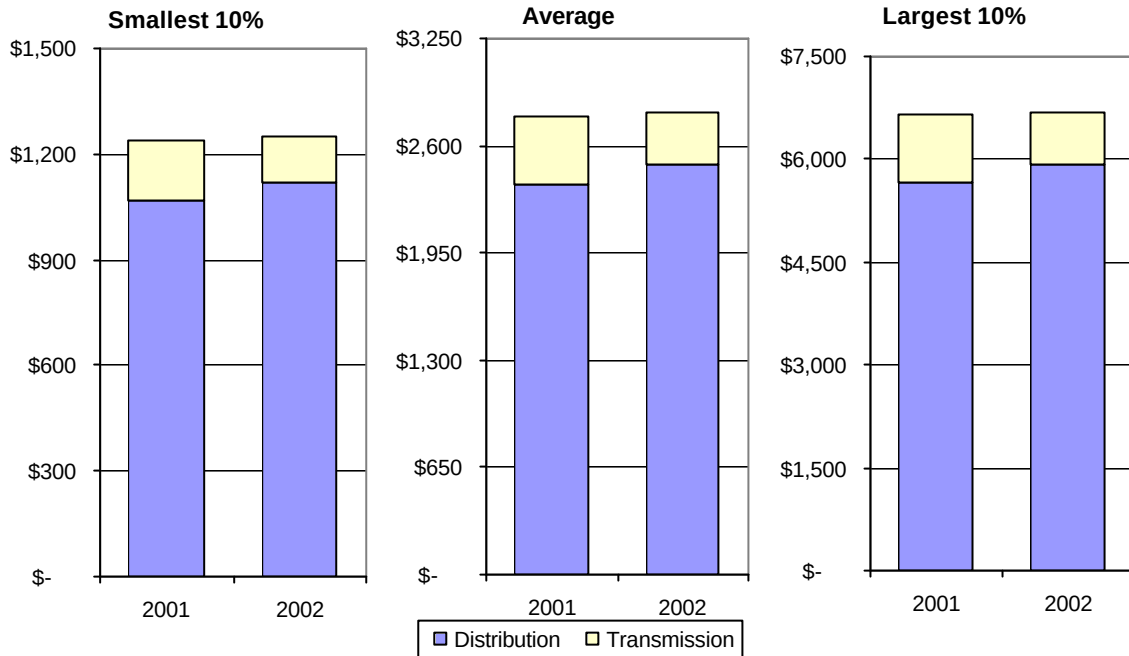


Fig 9: General Purpose Two Rate (5 day peak) – Average Annual Bill by Customer Segment)

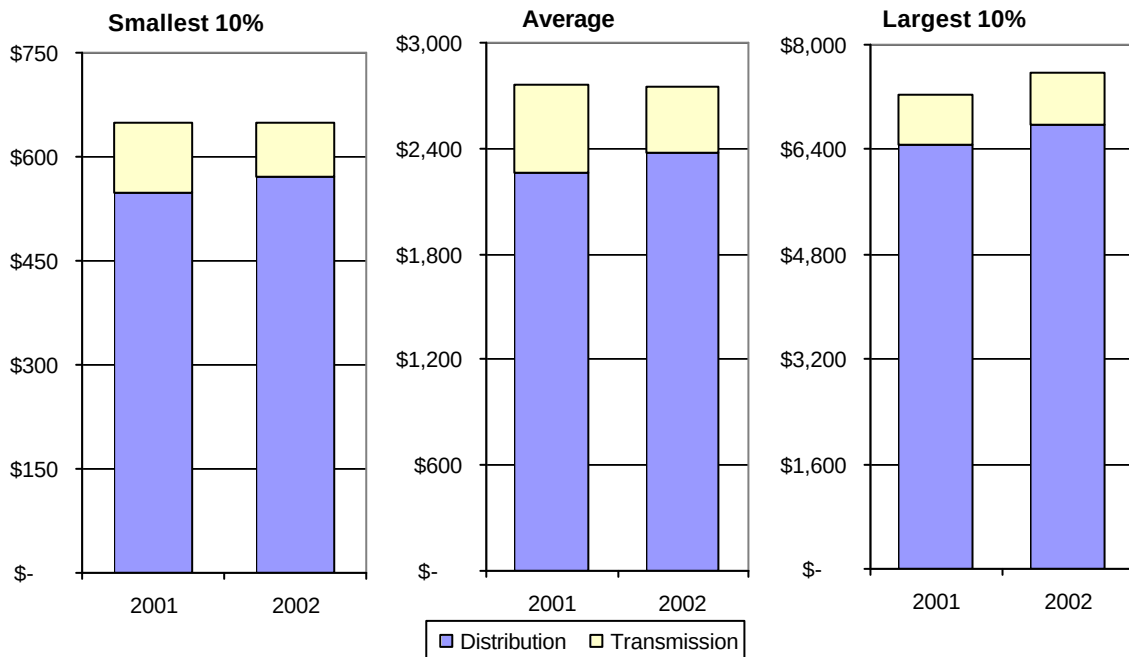


Fig 10: General Purpose Two Rate (5 day peak) – Bulk – Average Annual Bill by Customer Segment

5.1.6 General Purpose Two Rate (7 day peak)

These tariffs were based on the obsolete MUT E1, and are available only to existing installations supplied under this tariff. Existing customers are encouraged to consider the two rate, 5 day peak (weekends off-peak) or maximum demand tariffs as an alternative because they promote more efficient utilisation of the network by providing an appropriate time of use signal.

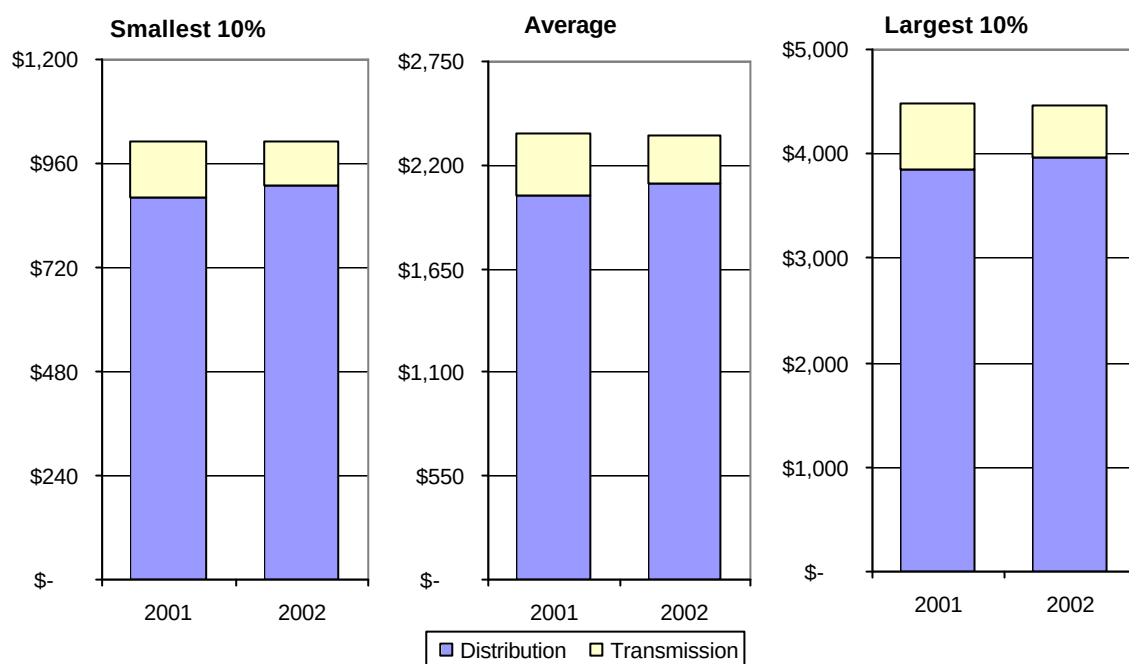


Fig 11: General Purpose Two Rate (7 day peak) – Average Annual Bill by Customer Segment

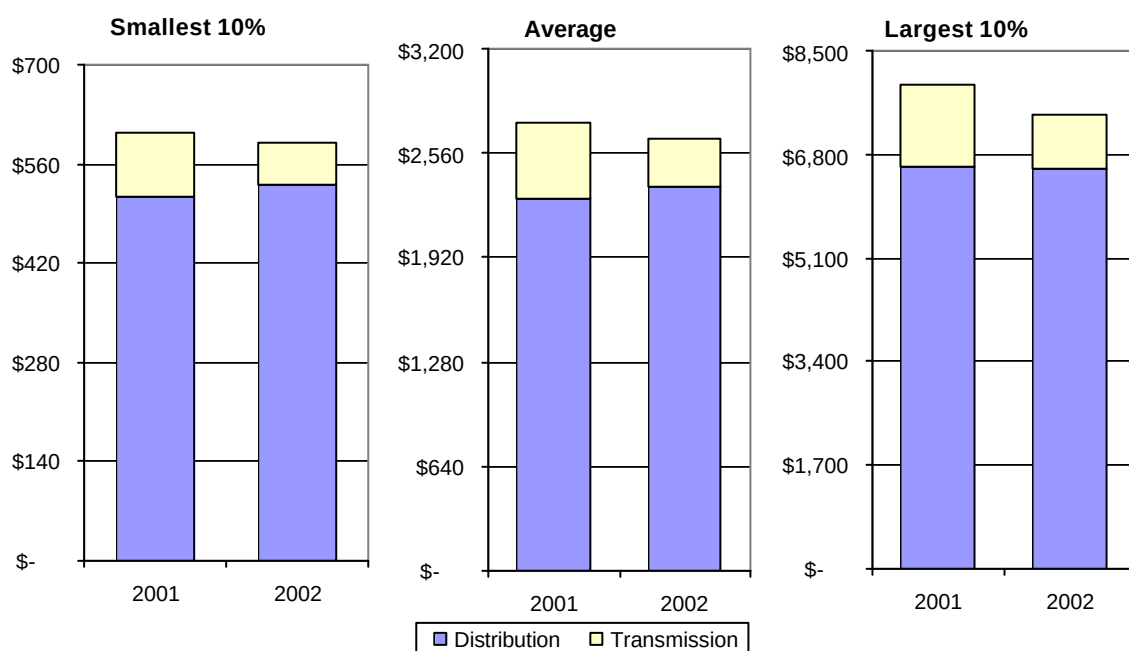


Fig 12: General Purpose Two Rate (7 day peak) Bulk – Average Annual Bill by Customer Segment

5.1.7 Large Two Rate (7 day peak)

This tariff is based on the obsolete MUT E1 and is available only to existing installations supplied under this tariff. There are relatively few customers supplied under this tariff and they are encouraged to consider the Large LV Demand metered tariff as an alternative because it promotes more efficient utilisation of the network by providing an appropriate time of use signal.

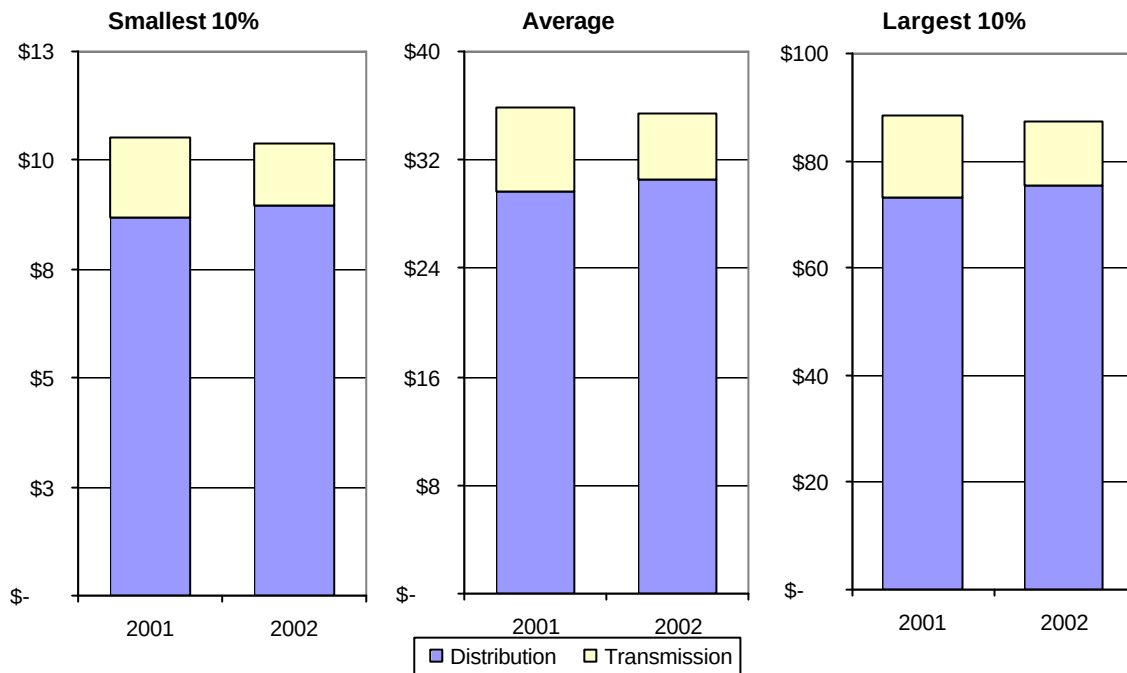


Fig 13: Large Two Rate (7 day peak) – Average Annual Bill (thousands of dollars) by Customer Segment

5.1.8 Large LV Demand Metered

This tariff is designed for larger commercial and industrial customers and provides both a demand signal and a time of use signal to promote efficient network use.

Currently the calculation of the demand component is based on a “contract demand” which locks in peak demands for at least one billing year. This contracted level of demand is fixed.

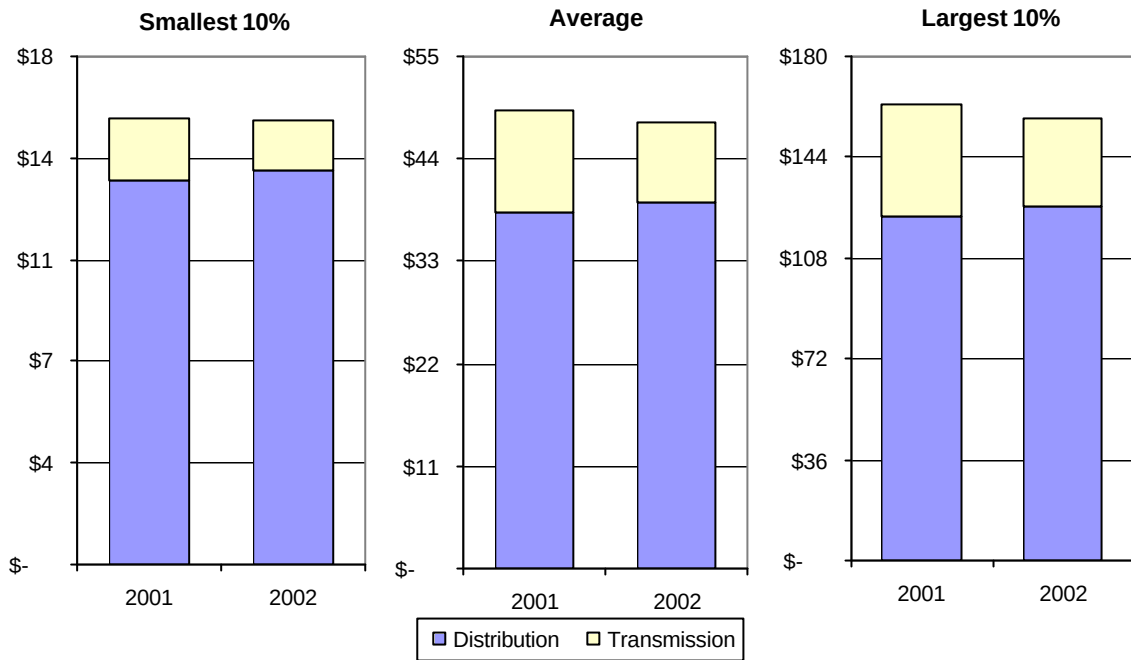


Fig 14: Large LV Demand Metered (Bulk) – Average Annual Bill (thousands of dollars) by Customer Segment

5.1.9 HV Demand Metered

This tariff is designed for larger commercial and industrial customers who require high voltage supply. It provides both a demand signal and a time of use signal to promote efficient network use. As with the Large LV Demand Metered Tariff, the calculation of the demand component is based on a “contract demand”.

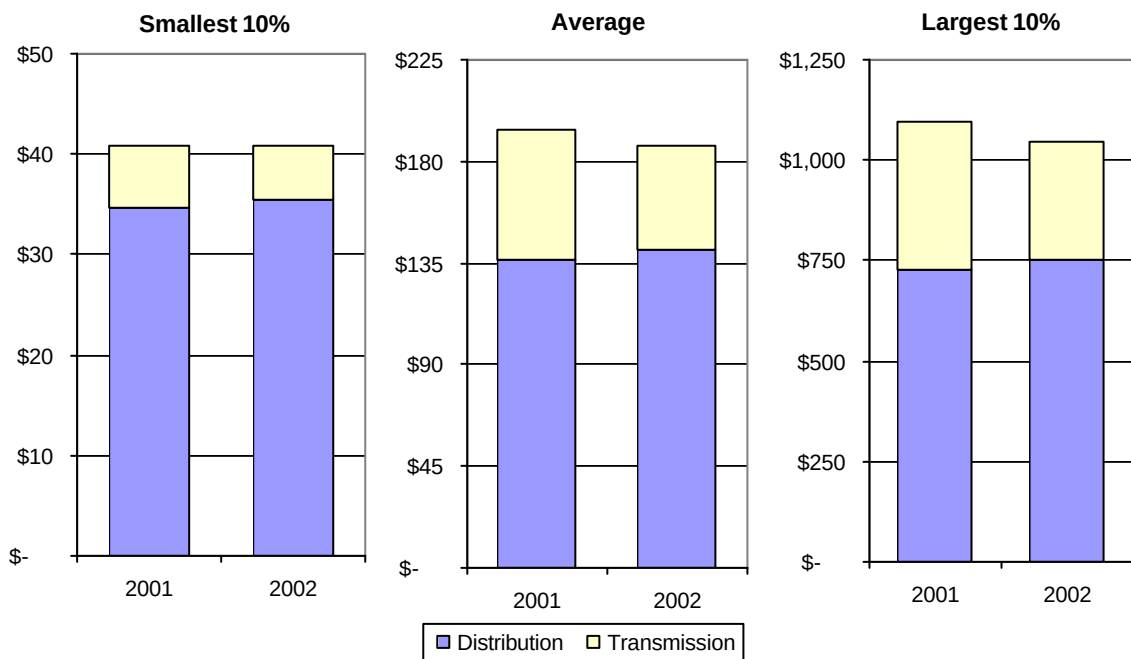


Fig 15: HV Demand Metered – Average Annual Bill (thousands of dollars) by Customer Segment

5.1.10 Subtransmission

This tariff applies only to a very limited number of large connections operating at 22,000 volts. It is heavily dominated by the transmission component. The significant reduction in the transmission component of the network tariff, illustrated above, reflect the impact of higher than expected settlement residues in reducing transmission charges. Should settlement residues diminish these reductions will may not be maintained in future years.

5.1.11 Unmetered Supplies

This tariff applies only to those situations where it is not economic to install a meter such as public lighting, telephone cabinets etc.

APPENDIX 1

NETWORK TARIFFS To Apply from 1 January 2002 EXCLUDING GST

Low Voltage Categories Nominal Voltage < 1000 volts	Component	Units	Price
Residential	Single Rate		
	Standing Charge:	\$ pa	21.19
	Unit charge:		
	First 1020 kWh per quarter	c/kWh	4.114
	Balance	c/kWh	5.367
	Single Rate - Bulk		
	Standing Charge:	\$ pa	18.28
	Unit charge:		
	First 1020 kWh per quarter	c/kWh	3.493
	Balance	c/kWh	4.389
	Two Rate		
	Standing Charge:	\$ pa	47.33
	Peak Unit charge:	c/kWh	5.638
	Off Peak Unit charge:	c/kWh	1.295
	Two Rate - Bulk		
	Standing Charge:	\$ pa	46.65
	Peak Unit charge:	c/kWh	4.791
	Off Peak Unit charge:	c/kWh	1.102
	Dedicated supplies		
	Standing Charge:	\$ pa	27.33
	Off Peak unit rate:	c/kWh	0.890
	Dedicated supplies - Bulk		
	Standing Charge:		
	Off Peak unit rate	\$ pa c/kWh	27.01 0.778

APPENDIX 1

NETWORK TARIFFS To Apply from 1 January 2002 EXCLUDING GST

General Purpose	Single rate		
	Standing Charge: Unit charge:	\$ pa c/kWh	47.74 5.192
	Single rate - Bulk		
	Standing Charge: Unit charge	\$ pa c/kWh	40.72 4.406
	Two rate (5 day peak)		
	Standing Charge: Peak unit rate: Off Peak unit rate:	\$ pa c/kWh c/kWh	113.40 6.297 1.430
	Two rate - Bulk (5 day peak)		
	Standing Charge: Peak unit rate: Off Peak unit rate:	\$ pa c/kWh c/kWh	105.72 5.135 1.145
	Two rate (7 day peak)		
	Standing Charge: Peak unit rate: Off Peak unit rate:	\$ pa c/kWh c/kWh	102.78 5.252 1.421
	Two rate - Bulk (7 day peak)		
	Standing Charge: Peak unit rate: Off Peak unit rate:	\$ pa c/kWh c/kWh	94.63 4.269 1.136
Unmetered Supplies (P/L etc)	Peak unit rate	c/kWh	5.538
	Off Peak unit rate	c/kWh	1.470

APPENDIX 1

NETWORK TARIFFS To Apply from 1 January 2002 EXCLUDING GST

Low Voltage Categories Nominal Voltage < 1000 volts	Component	Units	Price
Large Low Voltage	Demand Metered Standing Charge Peak unit rate Off Peak unit rate Demand charge Minimum chargeable demand	\$ pa c/kWh c/kWh \$/kW pa	3743.23 1.449 0.950 64.58 120 kW
	Demand Metered - Bulk Standing Charge Peak unit rate Off Peak unit rate Demand charge Minimum chargeable demand	\$ pa c/kWh c/kWh \$/kW pa	3743.23 1.189 0.930 59.42 120 kW
	Two rate (7 day peak) Standing Charge Peak unit rate Off Peak unit rate	\$ pa c/kWh c/kWh	99.47 4.811 1.376

High Voltage Tariffs Nominal Voltage ≥ 1000 volts and ≤ 22,000 volts	Component	Units	Price
High Voltage	Demand Metered Standing Charge Peak unit rate Off Peak unit rate Demand charge Minimum chargeable demand	\$ pa c/kWh c/kWh \$/kW pa	2370.82 1.117 0.611 46.32 1000 kW

Sub-Transmission Tariff Nominal voltage = 22,000 volts	Component	Units	Price
Sub-Transmission (22kV)	Standing Charge Peak unit rate Off Peak unit rate Demand charge Minimum chargeable demand.	\$ pa c/kWh c/kWh \$/kW pa	2461.40 1.086 0.281 11.69 10,000 kW

APPENDIX 1

DISTRIBUTION TARIFFS To Apply from 1 January 2002 EXCLUDING GST

Low Voltage Categories Nominal Voltage < 1000 volts	Component	Units	Price
Residential	Single Rate		
	Standing Charge:	\$ pa	21.19
	Unit charge:		
	First 1020 kWh per quarter	c/kWh	3.752
	Balance	c/kWh	5.005
	Single Rate - Bulk		
	Standing Charge:	\$ pa	18.28
	Unit charge:		
	First 1020 kWh per quarter	c/kWh	3.131
	Balance	c/kWh	4.027
	Two Rate		
	Standing Charge:	\$ pa	47.33
	Peak Unit charge:	c/kWh	5.002
	Off Peak Unit charge:	c/kWh	1.150
	Two Rate - Bulk		
	Standing Charge:	\$ pa	46.65
	Peak Unit charge:	c/kWh	4.155
	Off Peak Unit charge:	c/kWh	0.957
	Dedicated supplies		
	Standing Charge:	\$ pa	27.33
	Off Peak unit rate:	c/kWh	0.745
	Dedicated supplies - Bulk		
	Standing Charge:		
	Off Peak unit rate	\$ pa c/kWh	27.01 0.633

APPENDIX 1

DISTRIBUTION TARIFFS To Apply from 1 January 2002 EXCLUDING GST

General Purpose	Single rate		
	Standing Charge: Unit charge:	\$ pa c/kWh	47.74 4.686
	Single rate - Bulk		
	Standing Charge: Unit charge	\$ pa c/kWh	40.72 3.900
	Two rate (5 day peak)		
	Standing Charge: Peak unit rate: Off Peak unit rate:	\$ pa c/kWh c/kWh	113.40 5.552 1.285
	Two rate - Bulk (5 day peak)		
	Standing Charge: Peak unit rate: Off Peak unit rate:	\$ pa c/kWh c/kWh	105.72 4.390 1.000
	Two rate (7 day peak)		
	Standing Charge: Peak unit rate: Off Peak unit rate:	\$ pa c/kWh c/kWh	102.78 4.658 1.276
	Two rate - Bulk (7 day peak)		
	Standing Charge: Peak unit rate: Off Peak unit rate:	\$ pa c/kWh c/kWh	94.63 3.675 0.991
Unmetered Supplies (P/L etc)	Peak unit rate	c/kWh	5.392
	Off Peak unit rate	c/kWh	1.325

APPENDIX 1

DISTRIBUTION TARIFFS To Apply from 1 January 2002 EXCLUDING GST

Low Voltage Categories Nominal Voltage < 1000 volts	Component	Units	Price
Large Low Voltage	Demand Metered Standing Charge Peak unit rate Off Peak unit rate Demand charge Minimum chargeable demand	\$ pa c/kWh c/kWh \$/kW pa	3743.23 0.815 0.805 60.54 120 kW
	Demand Metered - Bulk Standing Charge Peak unit rate Off Peak unit rate Demand charge Minimum chargeable demand	\$ pa c/kWh c/kWh \$/kW pa	3743.23 0.555 0.785 55.38 120 kW
	Two rate (7 day peak) Standing Charge Peak unit rate Off Peak unit rate	\$ pa c/kWh c/kWh	99.47 4.134 1.231

High Voltage Tariffs Nominal Voltage ≥ 1000 volts and ≤ 22,000 volts	Component	Units	Price
High Voltage	Demand Metered Standing Charge Peak unit rate Off Peak unit rate Demand charge Minimum chargeable demand	\$ pa c/kWh c/kWh \$/kW pa	2370.82 0.376 0.466 43.37 1000 kW

Sub-Transmission Tariff Nominal voltage = 22,000 volts	Component	Units	Price
Sub-Transmission (22kV)	Standing Charge Peak unit rate Off Peak unit rate Demand charge Minimum chargeable demand.	\$ pa c/kWh c/kWh \$/kW pa	2461.40 0.100 0.136 10.69 10,000 kW

APPENDIX 1

TRANSMISSION TARIFFS To Apply from 1 January 2002 EXCLUDING GST

Low Voltage Categories Nominal Voltage < 1000 volts	Component	Units	Price
Residential	Single Rate		
	Standing Charge:	\$ pa	0.00
	Unit charge:		
	First 1020 kWh per quarter	c/kWh	0.362
	Balance	c/kWh	0.362
	Single Rate - Bulk		
	Standing Charge:	\$ pa	0.00
	Unit charge:		
	First 1020 kWh per quarter	c/kWh	0.362
	Balance	c/kWh	0.362
	Two Rate		
	Standing Charge:	\$ pa	0.00
	Peak Unit charge:	c/kWh	0.636
	Off Peak Unit charge:	c/kWh	0.145
	Two Rate - Bulk		
	Standing Charge:	\$ pa	0.00
	Peak Unit charge:	c/kWh	0.636
	Off Peak Unit charge:	c/kWh	0.145
	Dedicated supplies		
	Standing Charge:	\$ pa	0.00
	Off Peak unit rate:	c/kWh	0.145
	Dedicated supplies - Bulk		
	Standing Charge:		
	Off Peak unit rate	\$ pa c/kWh	0.00 0.145

APPENDIX 1

TRANSMISSION TARIFFS To Apply from 1 January 2002 EXCLUDING GST

General Purpose	Single rate Standing Charge: Unit charge:	\$ pa c/kWh	0.00 0.506
	Single rate - Bulk Standing Charge: Unit charge:	\$ pa c/kWh	0.00 0.506
	Two rate (5 day peak) Standing Charge: Peak unit rate: Off Peak unit rate:	\$ pa c/kWh c/kWh	0.00 0.745 0.145
	Two rate - Bulk (5 day peak) Standing Charge: Peak unit rate: Off Peak unit rate:	\$ pa c/kWh c/kWh	0.00 0.745 0.145
	Two rate (7 day peak) Standing Charge: Peak unit rate: Off Peak unit rate:	\$ pa c/kWh c/kWh	0.00 0.594 0.145
	Two rate - Bulk (7 day peak) Standing Charge: Peak unit rate: Off Peak unit rate:	\$ pa c/kWh c/kWh	0.00 0.594 0.145
Unmetered Supplies (P/L etc)	Peak unit rate	c/kWh	0.146
	Off Peak unit rate	c/kWh	0.145

APPENDIX 1

TRANSMISSION TARIFFS To Apply from 1 January 2002 EXCLUDING GST

Low Voltage Categories Nominal Voltage < 1000 volts	Component	Units	Price
Large Low Voltage	Demand Metered Standing Charge Peak unit rate Off Peak unit rate Demand charge Minimum chargeable demand	\$ pa c/kWh c/kWh \$/kW pa	0.00 0.634 0.145 4.04 120 kW
	Demand Metered - Bulk Standing Charge Peak unit rate Off Peak unit rate Demand charge Minimum chargeable demand	\$ pa c/kWh c/kWh \$/kW pa	0.00 0.634 0.145 4.04 120 kW
	Two rate (7 day peak) Standing Charge Peak unit rate Off Peak unit rate	\$ pa c/kWh c/kWh	0.00 0.677 0.145

High Voltage Tariffs Nominal Voltage ≥ 1000 volts and ≤ 22,000 volts	Component	Units	Price
High Voltage	Demand Metered Standing Charge Peak unit rate Off Peak unit rate Demand charge Minimum chargeable demand	\$ pa c/kWh c/kWh \$/kW pa	0.00 0.741 0.145 2.95 1000 kW

Sub-Transmission Tariff Nominal voltage = 22,000 volts	Component	Units	Price
Sub-Transmission (22kV)	Standing Charge Peak unit rate Off Peak unit rate Demand charge Minimum chargeable demand.	\$ pa c/kWh c/kWh \$/kW pa	0.00 0.986 0.145 1.00 10,000 kW